

Python Lists, Tuples & Dictionaries Quiz

1. What is the purpose of a try-except block?

- A) To prevent errors from occurring
- B) To handle errors gracefully
- C) To make code run faster
- D) To skip problematic code

Correct Answer: B

Explanation: try-except catches and handles exceptions to avoid crashes.

2. Which exception is raised for invalid type operations?

- A) ValueError
- B) SyntaxError
- C) TypeError
- D) NameError

Correct Answer: C

Explanation: Example: 'a' + 1 raises TypeError.

3. What does ZeroDivisionError indicate?

- A) Dividing zero by a number
- B) Dividing a number by zero
- C) Using zero as a list index
- D) Invalid zero assignment

Correct Answer: B

Explanation: Occurs in $x / 0$ or $x \% 0$.

4. Which is a base class for all exceptions?

- A) Exception
- B) BaseError
- C) BaseException
- D) Error

Correct Answer: C

Explanation: BaseException is the root class (includes SystemExit, KeyboardInterrupt).

5. What happens if an exception is unhandled?

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- A) The program continues silently
- B) The program crashes with a traceback
- C) Python ignores it
- D) The OS handles it

Correct Answer: B

Explanation: Unhandled exceptions terminate the program.

6. When does the else block execute in try-except?

- A) When an exception occurs
- B) When no exceptions occur
- C) Always
- D) Only for specific exceptions

Correct Answer: B

Explanation: else runs if try succeeds.

7. What is the purpose of finally?

- A) To skip error handling
- B) To execute code regardless of exceptions
- C) To replace else
- D) To ignore all errors

Correct Answer: B

Explanation: Example: Cleanup code (e.g., closing files).

8. What is the output of:

```
try:  
    print(1 / 0)  
except ZeroDivisionError:  
    print("A")  
else:  
    print("B")  
finally:  
    print("C")
```

- A) A B C

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B) A C

C) B C

D) C

Correct Answer: B

Explanation: else is skipped due to the exception; finally always runs.

9. Which is syntactically correct?

A) try: ... except: ... else: ...

B) try: ... else: ... except: ...

C) try: ... except: ... finally: ...

D) try: ... finally: ... except: ...

Correct Answer: C

Explanation: Order must be try -> except -> else/finally.

10. What does this code do?

```
try:
```

```
    x = int("text")
```

```
except ValueError as e:
```

```
    print(f"Error: {e}")
```

A) Converts "text" to an integer

B) Catches invalid conversion and prints the error

C) Raises a new exception

D) Silently ignores the error

Correct Answer: B

Explanation: Catches ValueError and prints its message.

11. Which block is optional in try-except?

A) try

B) except

C) finally

D) All except try and at least one except/finally

Correct Answer: D

Explanation: else and finally are optional, but you need at least except or finally.

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12. How do you manually raise an exception?

- A) throw ValueError("Error")
- B) raise ValueError("Error")
- C) exception ValueError("Error")
- D) error ValueError("Error")

Correct Answer: B

Explanation: raise triggers exceptions explicitly.

13. What is the purpose of custom exceptions?

- A) To replace built-in exceptions
- B) To create application-specific errors
- C) To make code run faster
- D) To ignore all errors

Correct Answer: B

Explanation: Example: class MyError(Exception): pass

14. Which is true about assert?

- A) It catches exceptions
- B) It raises AssertionError if the condition is False
- C) It ignores errors
- D) It replaces try-except

Correct Answer: B

Explanation: Example: assert x > 0, "x must be positive".

15. What does this code do?

```
def divide(a, b):  
    if b == 0:  
        raise ValueError("Cannot divide by zero")  
    return a / b
```

- A) Silently handles division by zero
- B) Raises a descriptive error for division by zero
- C) Returns None for invalid input
- D) Converts b to a non-zero value

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Correct Answer: B

Explanation: Custom error messages improve debugging.

16. How do you create a custom exception with a message?

- A) `class MyError(Exception): pass`
- B) `class MyError(Exception): def __init__(self, msg): super().__init__(msg)`
- C) `class MyError: pass`
- D) `def MyError(): return "Error"`

Correct Answer: B

Explanation: Proper inheritance and `__init__` are needed for custom messages.

17. What does `except (ValueError, TypeError)` catch?

- A) Only the first occurring exception
- B) Both `ValueError` and `TypeError`
- C) All exceptions
- D) No exceptions

Correct Answer: B

Explanation: Parentheses group multiple exceptions.

18. How do you catch all exceptions?

- A) `except:`
- B) `except Exception:`
- C) `except BaseException:`
- D) Both A and B

Correct Answer: D

Explanation: Avoid bare `except:`-it catches even `KeyboardInterrupt`!

19. What is the output of:

```
try:
    print(1 / 0)
except ZeroDivisionError:
    print("A")
except ArithmeticError:
```

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`print("B")`

- A) A
- B) B
- C) A B
- D) Error

Correct Answer: A

Explanation: ZeroDivisionError is a subclass of ArithmeticError-first match wins.

20. Which is true about exception chaining?

- A) Exceptions cannot be chained
- B) Use raise ... from ... to link exceptions
- C) Chaining hides the original error
- D) Only works for custom exceptions

Correct Answer: B

Explanation: Example: raise ValueError("Invalid") from e